

Digital Twin of Power Converter Systems

Identification and Optimization for Reliability

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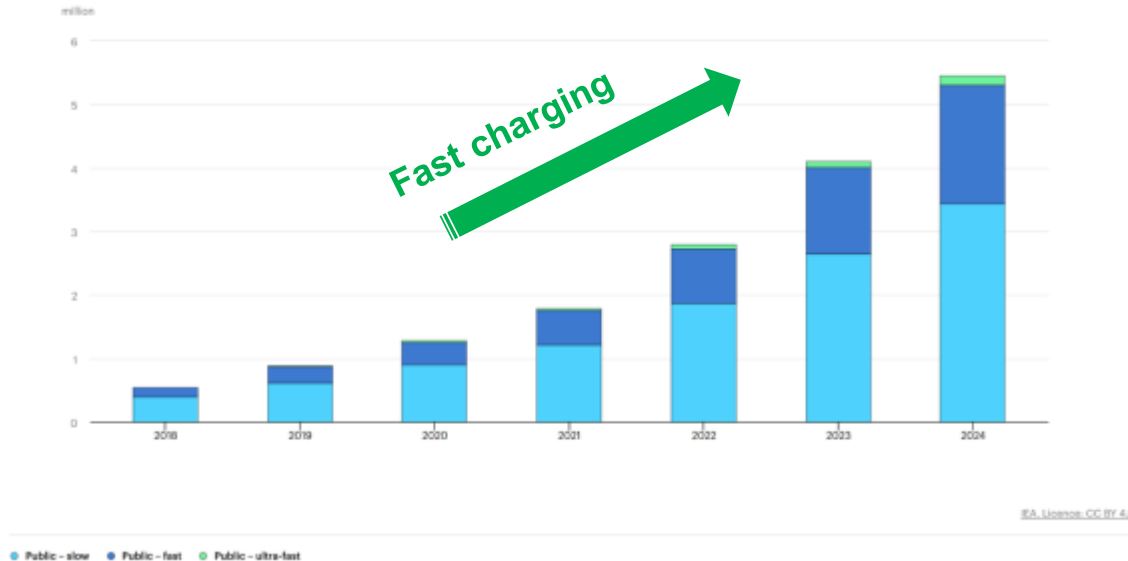
03-03-2025



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- Introduction & Motivation
- Review of Identification Methods
- Methodology
- Results
- Conclusion



EV charger Trends:

- Increase in EV chargers
- Higher power charger trend
- Increased converter usage
- Need for robust reliability and health monitoring.
- Need for accurate thermal models to maximize safe converter usage.
- Real time embedded deployment of monitoring

- ✓ The DT processes real-time sensor data (I, V, T) to adapt a twin model of a process.
- ✓ Thermal 'constants' drift due to aging; the Digital Twin adaptively mirrors the physical module's degradation.

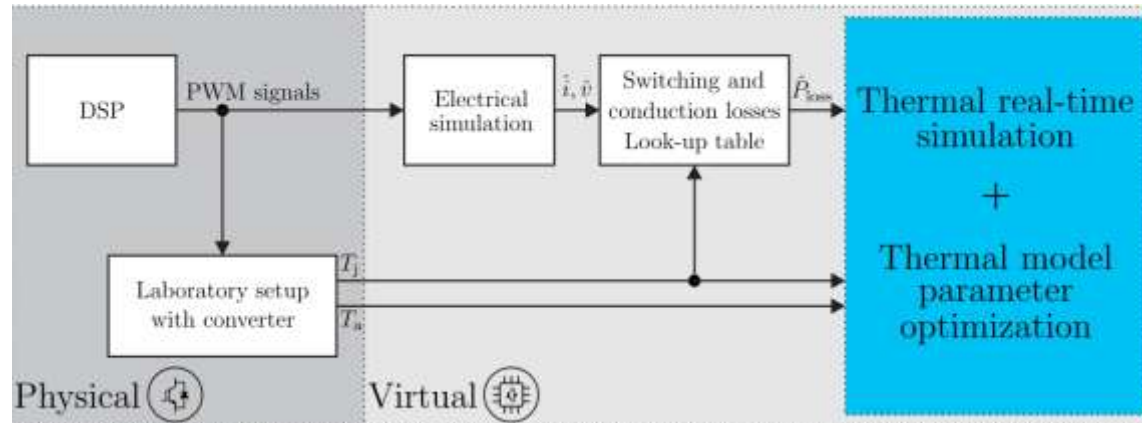


Fig.2 Digital twin Topology [2]

Common modeling approaches include Lumped Parameter Thermal Networks (LPTN), FEA, and Data-Driven models.

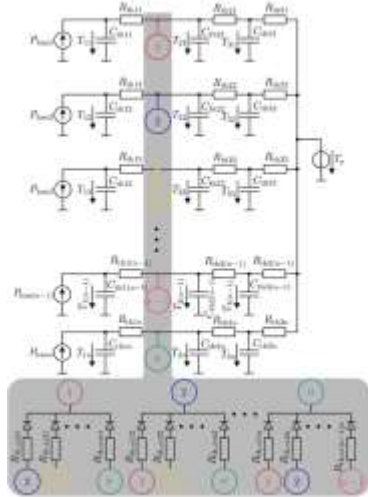


Fig.3 Cauer Network [2]

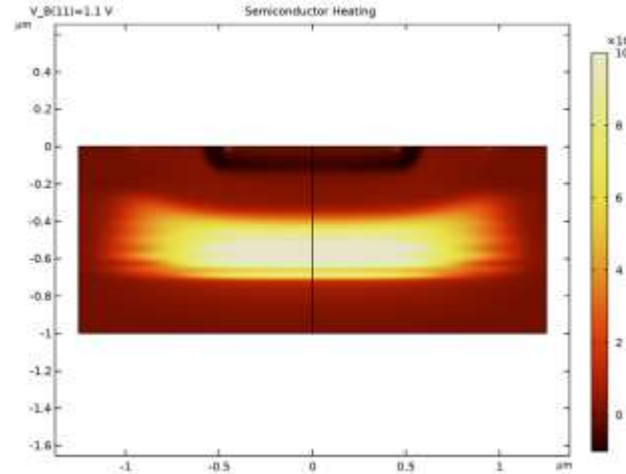


Fig.4 Semiconductor Heating FEA[3]

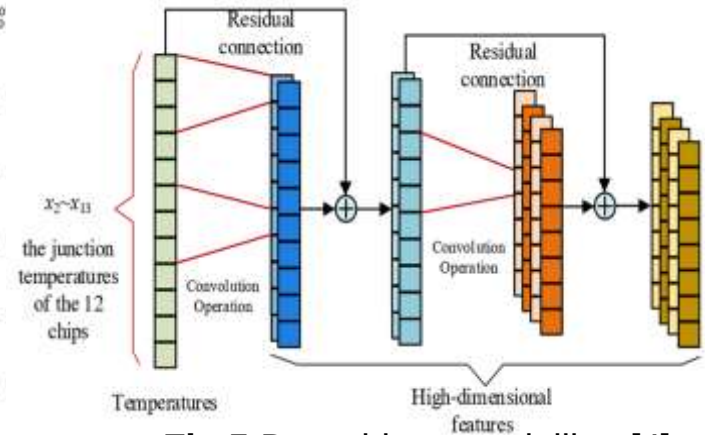


Fig.5 Data driven modelling [4]

Monitoring Principle

- **Network layers** represent physical layers.
- **Node Temperatures** represent average physical temperatures
- **Variation in thermal parameters** correlated with physical degradation.

Advantages

- ✓ Easy Implementation in embedded systems
- ✓ Computationally light
- ✓ Easy interpretation

Disadvantages

- r Over-simplifies dynamics
- r Parameters are not measurable
- r Thermal parameters are fixed

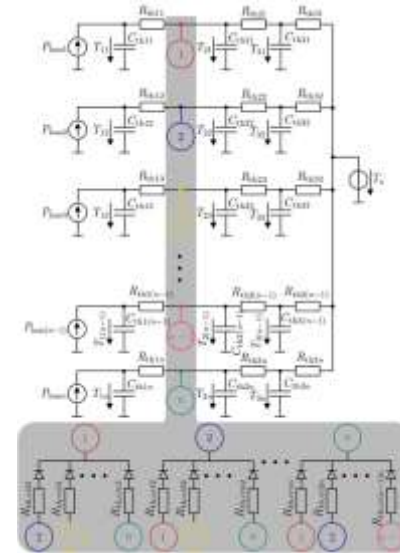


Fig.3 Cauer Network [2]

A reliable computationally light identification method is required

Experimental Validation

Experimental Validation Strategy: EKF, PSO and a data driven approach will be analyze for feasibility

- **(PSO & EKF):** Validated on **single-chip** self-heating data to identify localized Cauer parameters.
- **Data-Driven (H-DMD):** Validated on a **dual-chip** setup with cycling power to capture complex thermal cross-coupling.

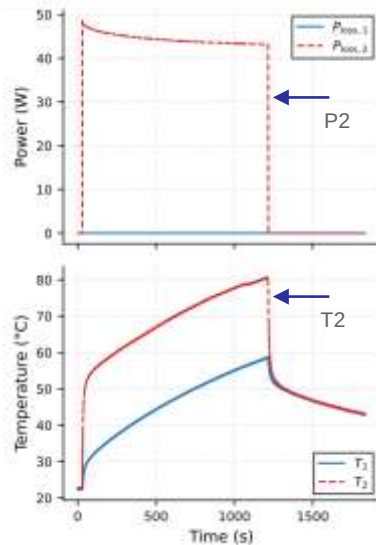


Fig. 6 Experiment A self heat [†]

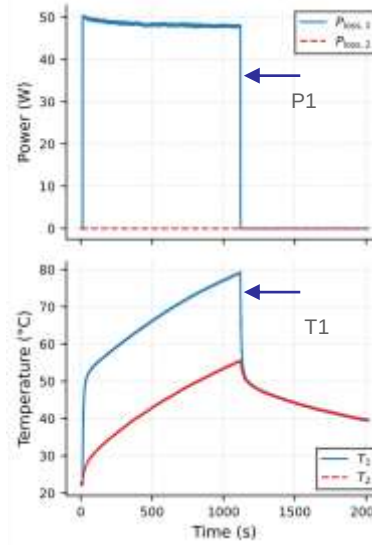


Fig. 7 Experiment B self heat [†]

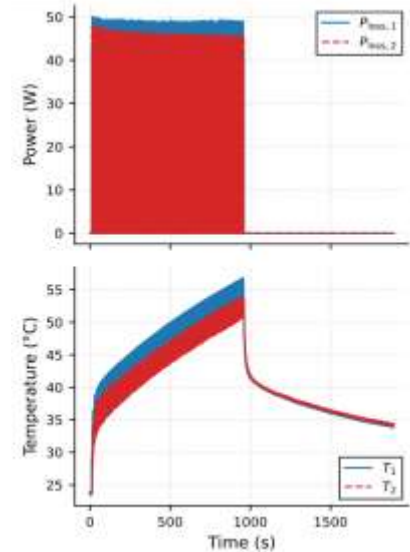


Fig. 8 Experiment C cycling [†]

Augmented EKF for Joint State and Log-Parameter Estimation

The Augmented Model

- The state space is expanded:

$$\mathbf{x}_{aug} = [T, T_{ref}, \dots, \ln(R_{th,1}), \dots, \ln(C_{th,2})]^T$$

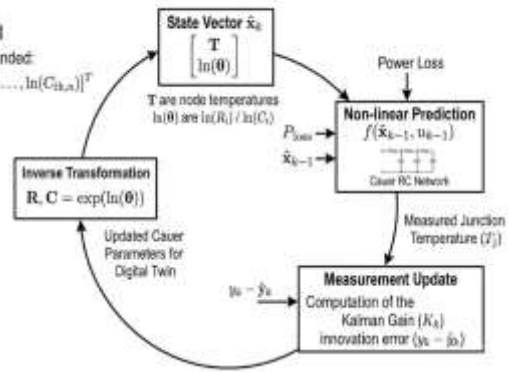


Fig. 9 EKF scheme

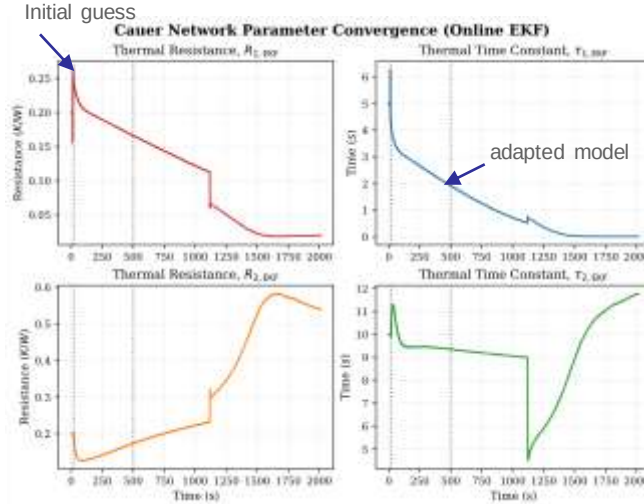


Fig. 10 EKF convergence

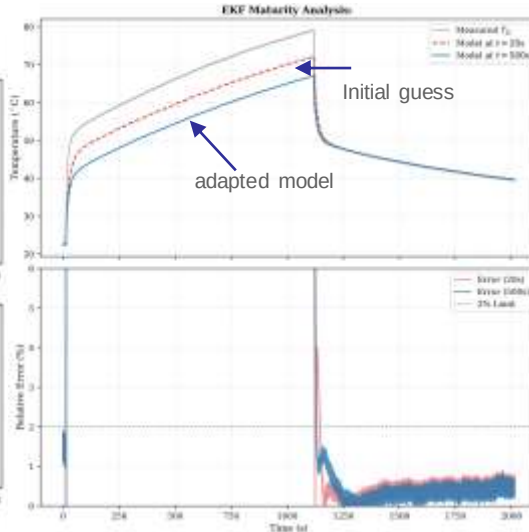
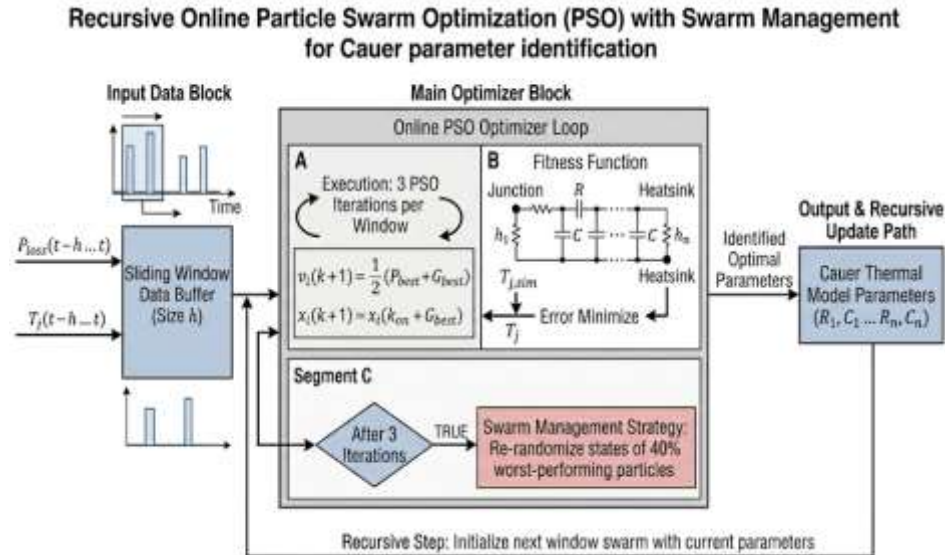


Fig. 11 EKF validation

- Good for one step prediction
- Can not separate the fast and slow dynamics

- **Global Identification:** Optimizes R_{th} and C_{th} without Jacobian requirements.
- **Real-time Constraint:** Execution limited to **3 iterations per window**.
- **Swarm Diversity:** **40% re-randomization** of worst-performing particles.
- **Recursive Warm-start**



PSO Identification Accuracy

- **Validation:** Tested on single-chip self-heating
- **Steady-State Precision:** Achieves during stable operating points
- **Transient Dynamics:** error "spikes" occur during high $\frac{dT}{dt}$ events
- **Convergence:** Rapid parameter stabilization within the first 3–5 windows

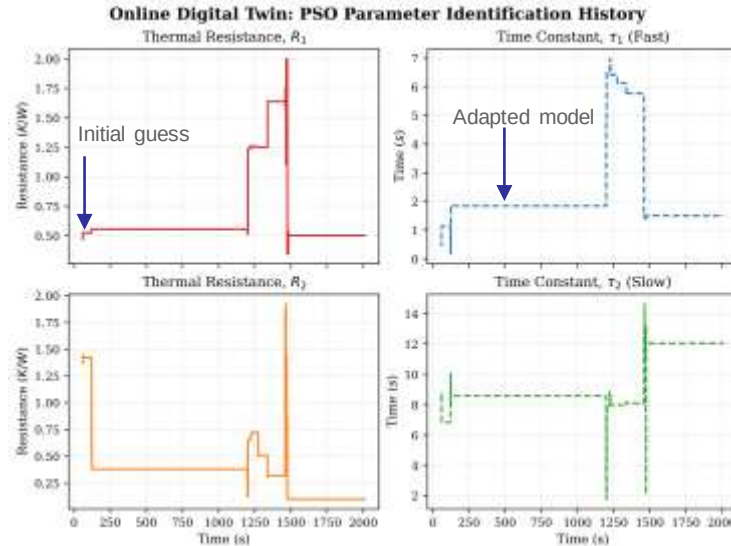


Fig. 13 PSO Convergence

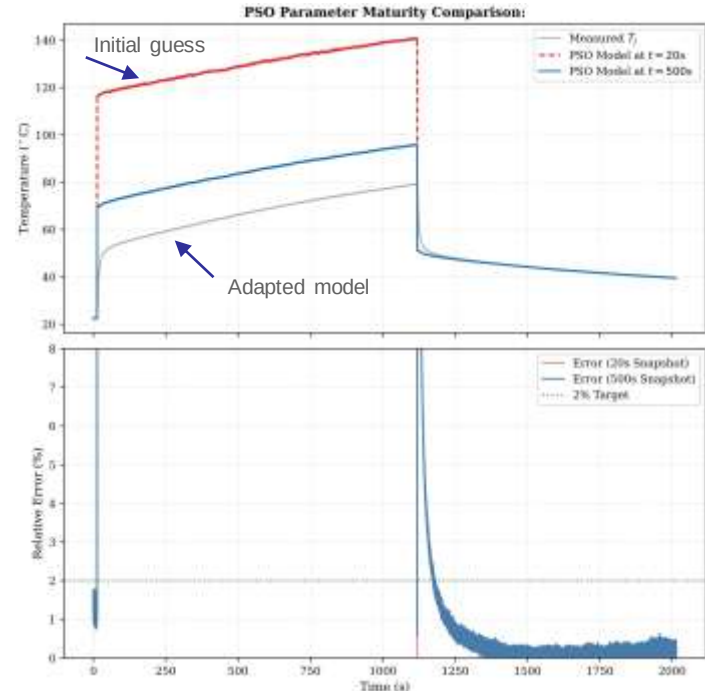


Fig. 14 PSO Validation

Goal: Define a system $T_{k+1} = G[T_k P_k]^T$

We need to find a matrix G that advances the state respecting the dynamics

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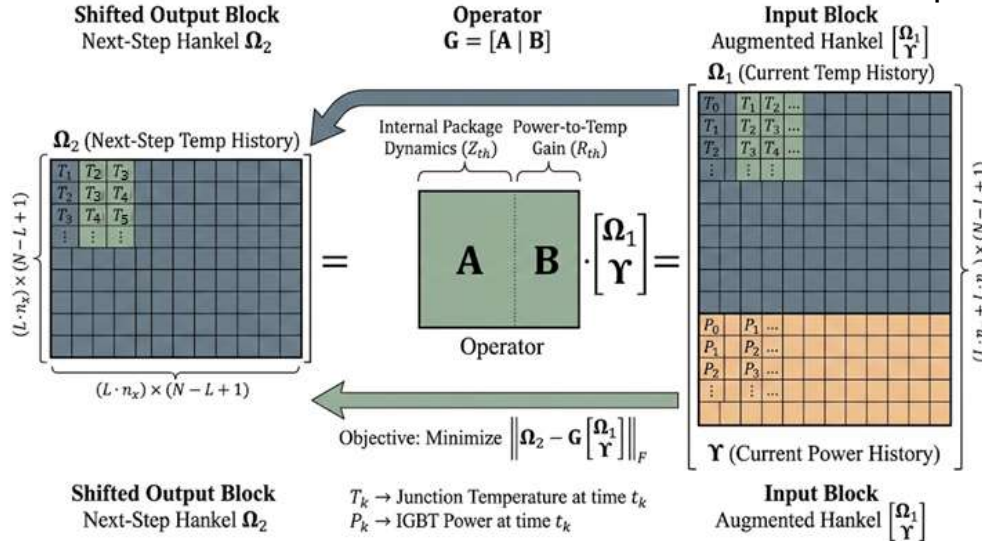


Fig. 15 Hankel DMD [*]

Goal: Define a system $T_{k+1} = G[T_k P_k]^T$

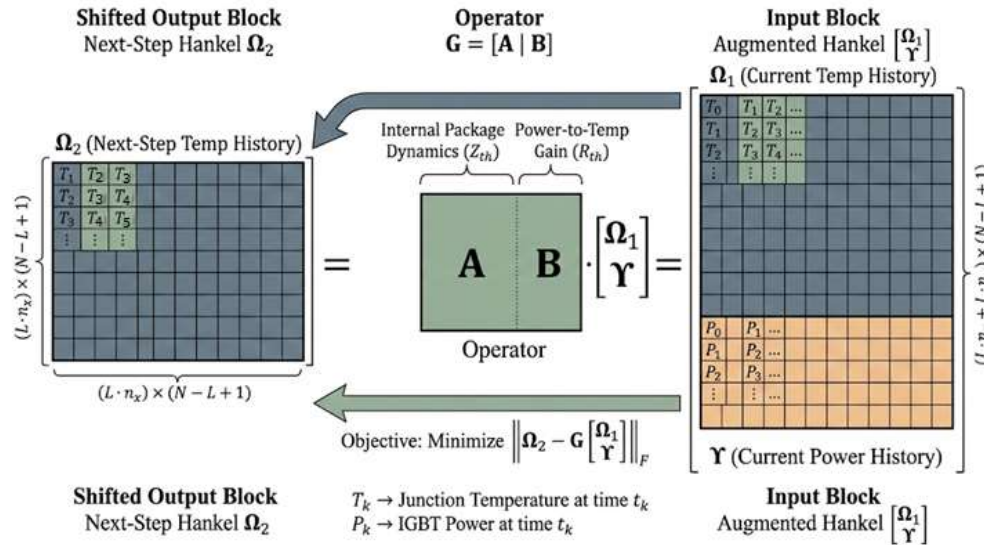


Fig. 15 Hankel DMD [*]

$G = \Omega_2 \Omega_{aug}^{-1} \rightarrow$ Computationally expensive

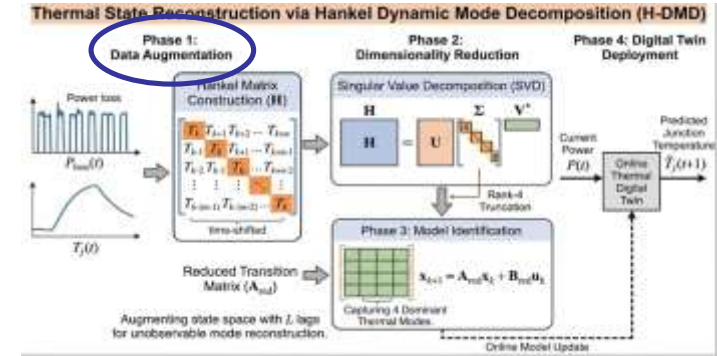


Fig. 16 DMD DT Scheme [*]

Goal: Define a system $T_{k+1} = G[T_k P_k]^T$

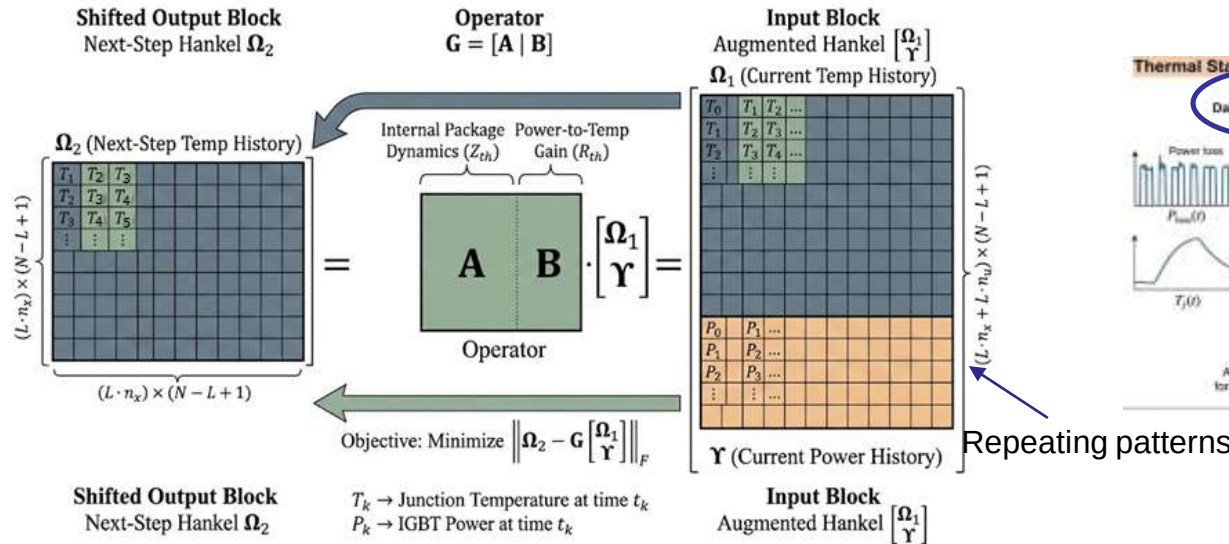


Fig. 15 Hankel DMD [*]

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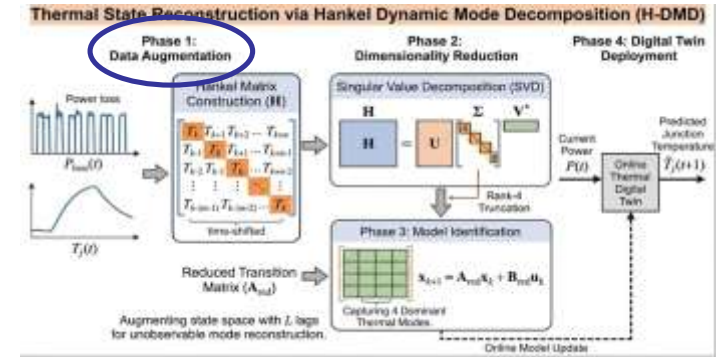


Fig. 16 DMD DT Scheme [*]

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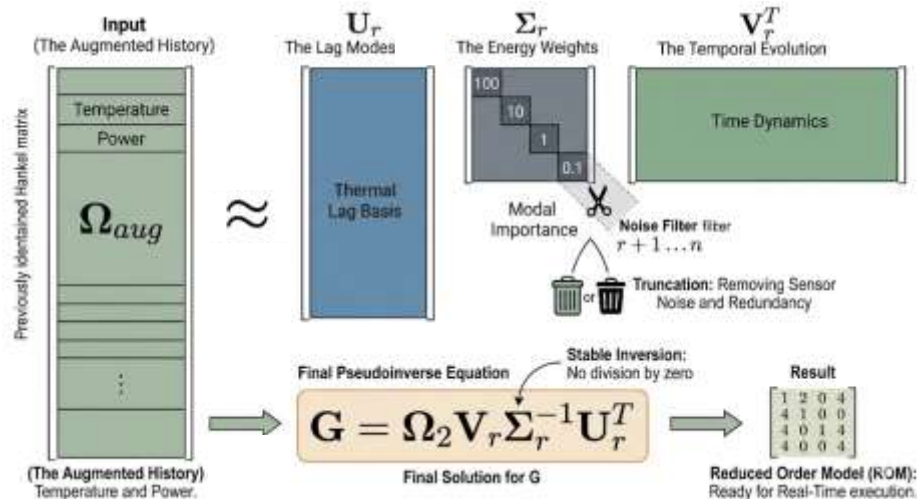


Fig.17 SVD psudo-inversion [*]

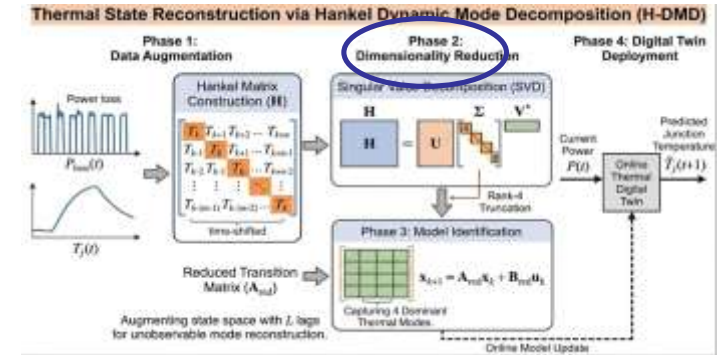


Fig. 16 DMD DT Scheme [*]

Goal: Adapt a system $T_{k+1} = G[T_k P_k]^T$

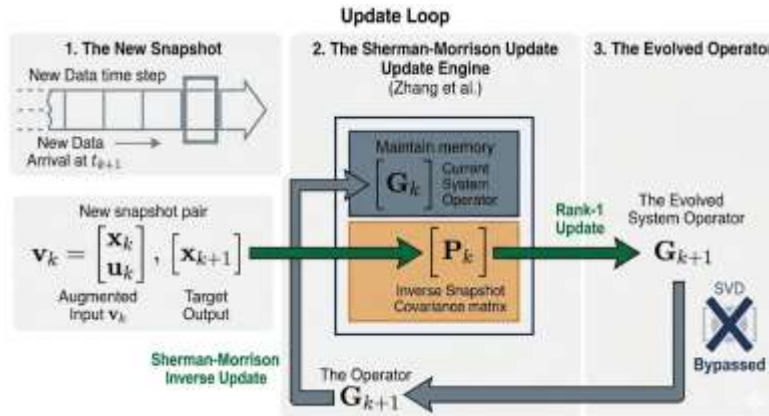


Fig. 18 Online DMD RLS [4]

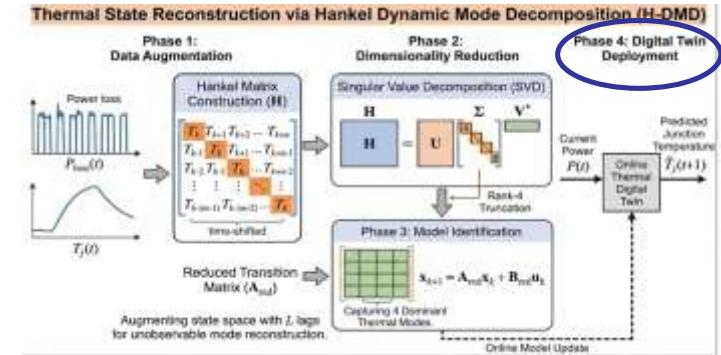


Fig. 16 DMD DT Scheme [4]

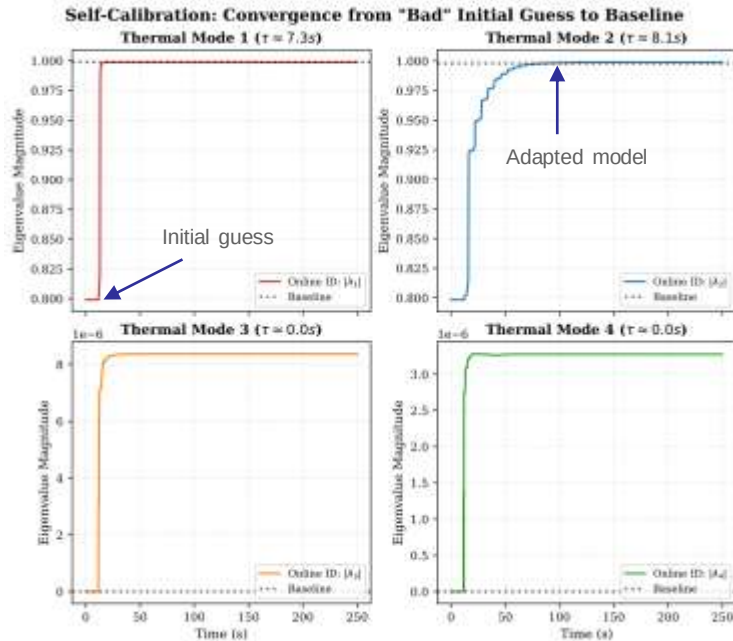


Fig. 19 Online DMD convergence

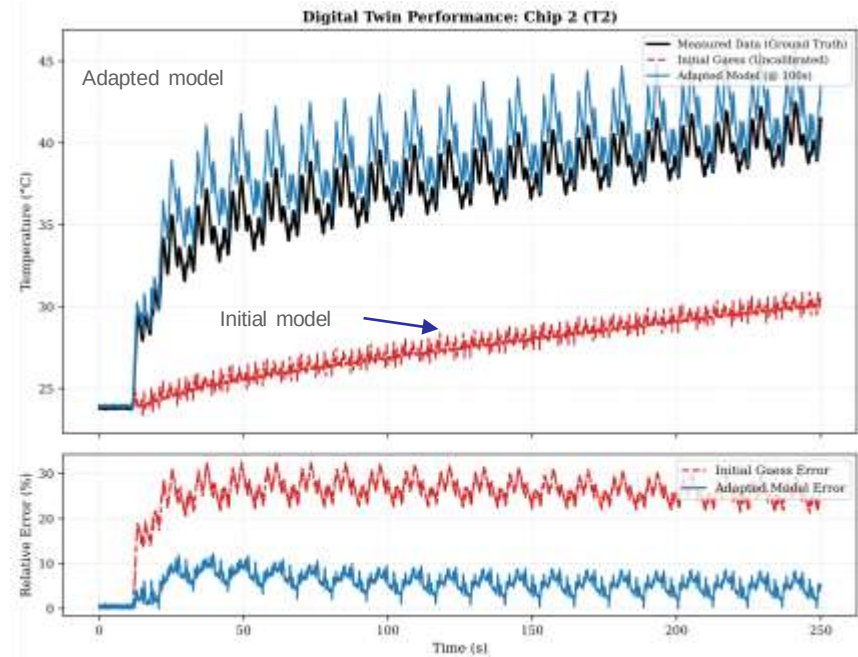


Fig. 20 DMD DT Validation

Summary

Method	Primary Strength	Main Challenge	Best Use Case
Online PSO (with 40% Reset)	Global Search: Excellent at finding parameters in non-convex spaces without needing a gradient/Jacobian.	Stochastic: Computationally heavier per iteration; requires windowed data for convergence.	Initial Calibration: Best for finding the baseline "Golden Model" Caue parameters offline or during startup.
Augmented EKF (Log-Space)	Recursive Efficiency: Updates at every sample; provides a covariance matrix (uncertainty) for health monitoring.	Local Convergence: Requires accurate initial guesses; sensitive to process noise tuning.	Continuous Tracking: Ideal for real-time monitoring of parameter drift (aging) during operation.
Hankel-DMD (Reduced Order)	Temporal Memory: Captures complex diffusion dynamics (4-mode) without needing a manual RC structure.	Data Driven: Lacks direct physical layer interpretability.	High-Speed Control: Best for fast Tj prediction in complex multi-chip modules with cross-heating.

Conclusion: The ideal method depends on the module complexity, and desired application, but it should be sensitive to both fast and slow thermal constants.

- ✓ Extend the Digital Twin to include heatsink temperature-based adaptation.
- ✓ Enhance the physical interpretability of DMD-derived thermal parameters.

Future Work: Adaptive H-DMDc Observer with Closed-Loop RLS-based Parameter Tracking

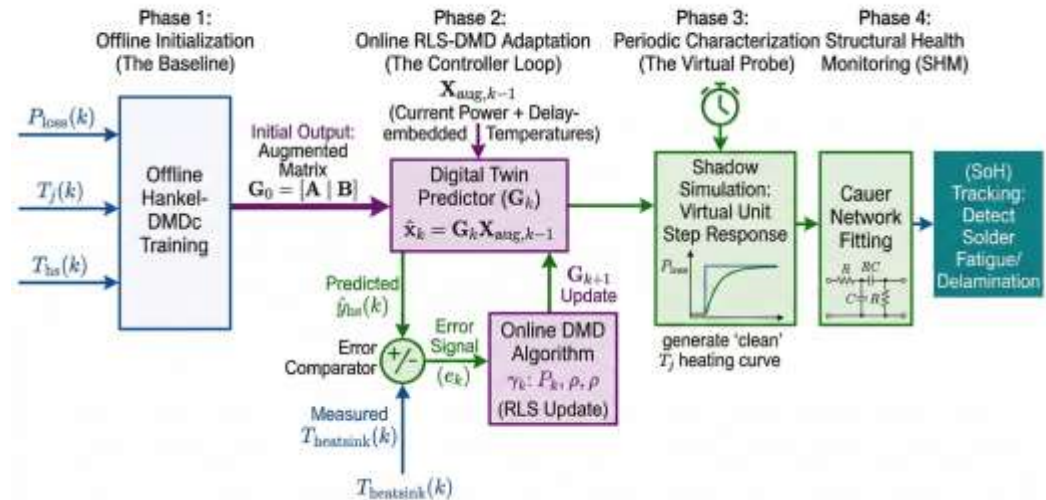


Fig. 20 Future work scheme [x]

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- [3] COMSOL, “Thermal Analysis of a Bipolar Transistor.” Accessed: Dec. 29, 2025. [Online]. Available: <https://www.comsol.com/model/thermal-analysis-of-a-bipolar-transistor-19701>
- [4] Y. Tang et al., “Physics-Informed Intelligent Thermal Model for SiC Power Modules With High Generalization,” IEEE Trans. Power Electron., vol. 40, no. 10, pp. 14459–14477, Oct. 2025. doi: 10.1109/TPEL.2025.3577241.
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[†] **Experimental Validation:** All raw thermal experimental data for the modules was graciously provided by Dr. Martin Votava

[*] **Technical Visualizations Note (diagrams):**

Some of the diagrams and system architecture illustrations presented in this seminar were designed by the author and rendered using Generative AI (Gemini 3 Flash/Nano Banana 2). All visual logic, mathematical notations, and block structures are derived directly from the underlying CauerEKF and DMDc Python implementation to ensure technical fidelity.